

Gluing one almost-everywhere constant over a countable cover

MathlibAnnex.aeConstantOn_of_countableCover

theorem

Combines local exceptional sets using an inequality between restricted measures.

Statement

Let (X, μ) be a measure space, Y any set of values, $u : X \rightarrow Y$, and $c \in Y$. Let $U \subseteq X$, let $(V_i)_{i \in I}$ be any family of subsets, and let $T \subseteq I$ be countable. Suppose

$$U \subseteq \bigcup_{i \in T} V_i, \quad u = c \quad \mu|_{V_i}\text{-almost everywhere for every } i \in T.$$

Then $u = c$ almost everywhere for $\mu|_U$.

Assumptions

There is no topology on X or Y in this statement. The function u , the cover members V_i , and U need not be measurable. Only T is countable; the index set I need not be. All members use the same constant c .

Conclusion

The one exceptional set $\{x : u(x) \neq c\}$ has measure zero for $\mu|_U$. The result includes the empty cover: the covering hypothesis then forces $U = \emptyset$.

The notation $\mu|_V$ means restriction of a measure to a set in the library's sense, which is defined even for a nonmeasurable V . The proof uses inequalities of measures and does not replace every restricted evaluation by an intersection formula requiring measurability.

Proof route

Compare $\mu|_U$ with the sum of the restricted measures of the countable cover, and evaluate this inequality on the exceptional set.

▼ Proof steps

1. **Form a countable family of measures.** Reindex by $i \in T$ itself and write $B = \{x \in X : u(x) \neq c\}$.

Restriction is monotone in its restricting set, and restriction to a countable union is bounded by the sum of the restrictions. Thus

$$\mu|_U \leq \mu|_{\bigcup_{i \in T} V_i} \leq \sum_{i \in T} \mu|_{V_i}.$$

These are measure inequalities; the restriction-to-union inequality does not require measurable cover members.

2. **Evaluate on the exceptional set.** Almost-everywhere equality on V_i says $(\mu|_{V_i})(B) = 0$. Evaluation of a countable sum of measures on any set therefore gives

$$0 \leq (\mu|_U)(B) \leq \left(\sum_{i \in T} \mu|_{V_i} \right) (B) = \sum_{i \in T} (\mu|_{V_i})(B) = 0.$$

Hence $(\mu|_U)(B) = 0$, which is exactly the required almost-everywhere equality. No uncountable union of null sets enters this argument.

Main citations

- Exact declaration and its source — `MathlibAnnex.aeConstantOn_of_countableCover`
- Almost-everywhere equality on a restricted measure — `MathlibAnnex.AEConstantOn`

Lean source signature (exact)

```
theorem aeConstantOn_of_countableCover
  {U : Set X} {u : X → Y} {c : Y}
  {ι : Type*} {V : ι → Set X} {T : Set ι}
  (hT : T.Countable) (hcover : U ⊆ ⋃ i ∈ T, V i)
  (hae : ∀ i ∈ T, AEConstantOn μ u (V i) c) :
  AEConstantOn μ u U c
```

In the source	Mathematical meaning
<code>{X Y : Type*} [MeasurableSpace X] (μ : Measure X)</code>	The measure space (X, μ) and arbitrary set of values Y ; no topology or linear structure is needed.
<code>{U : Set X} {u : X → Y} {c : Y}</code>	The domain U , function u and one fixed value c .
<code>{ι : Type*} {V : ι → Set X} {T : Set ι} (hT : T.Countable)</code>	The family $(V_i)_{i \in I}$ and countable subset $T \subseteq I$ of selected indices. The whole index set need not be countable.
<code>(hcover : U ⊆ ⋃ i ∈ T, V i)</code>	The covering hypothesis $U \subseteq \bigcup_{i \in T} V_i$.
In the source	Mathematical meaning
<code>(hae : ∀ i ∈ T, AEConstantOn μ u (V i) c)</code>	For every selected i , $u = c$ almost everywhere for $\mu _{V_i}$, always with the same constant.
<code>AEConstantOn μ u U c</code>	The conclusion is $u = c$ almost everywhere for $\mu _U$. No measurability of u, U, V_i is silently imposed.

▼ Exact surrounding binder context (separate excerpts)

Exact source lines 12–18:

```
noncomputable section
open Set MeasureTheory Filter TopologicalSpace
open scoped ENNReal Topology
namespace MathlibAnnex

section Measurable
variable {X Y : Type*} [MeasurableSpace X] (μ : Measure X)
```

▼ Exact content identity

Declaration: MathlibAnnex.aeConstantOn_of_countableCover

Accepted content SHA-256: 7073f259f8ecabc04b8f2aca208c67c5742d15bea096432658b0aefe35a6057b

Accepted source guide SHA-256: edf9847bd4ae748ce15f59448e57dfea1082708eb4241091f282939ecb186b40

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73